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Executive Summary

Parl is a protocol for parimutuel prediction markets — a decentralized infrastructure that enables anyone to create, manage, and settle markets on any outcome without requiring liquidity providers, market makers, or order books.

Traditional prediction markets rely on automated market makers (AMMs) or order-book matching. Both face fundamental scalability constraints: AMMs require deep liquidity pools and expose liquidity providers to impermanent loss; order books fragment liquidity across individual markets and require constant market-maker participation.

Parimutuel betting — the same mechanism that powers horse racing pools worldwide — resolves these constraints elegantly. All bets on a given market pool into a single liquidity pot. The payout is determined algorithmically after the outcome is known: winners split the pool (minus protocol fee). There is no need for continuous price discovery, no order matching, and no liquidity provider risk.

Parl is infrastructure, not an application. The protocol provides:

1. A parimutuel pool engine
2. A resolution framework (oracle-agnostic)
3. An SDK for market creation and integration

Applications built on top of Parl can range from sports betting and political forecasting to enterprise prediction and community markets.

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1. Problem Statement

1.1 The Prediction Market Landscape

Current prediction market platforms fall into two broad categories:

1.1.1 Automated Market Maker (AMM) Based

Polymarket, *Augur*, and similar platforms use AMM curves (typically logarithmic or constant-product) to determine prices. Liquidity providers deposit tokens into pools and earn fees in exchange for providing liquidity.

Problems:

- **Liquidity fragmentation:** Each market requires a separate pool. Popular markets absorb liquidity; niche markets remain illiquid or nonviable.
- **Impermanent loss (IL):** LPs face divergence loss when market prices move, discouraging participation.
- **Capital inefficiency:** Deep pools are required even for small-volume markets to maintain reasonable slippage.
- **Complex bootstrapping:** New markets need external capital to seed liquidity before trading begins.

1.1.2 Order Book Based

Kalshi and traditional financial exchanges use continuous double auctions with market makers.

Problems:

- **Requires professional market makers:** Without dedicated MMs, spreads are wide and depth is thin.
- **Not permissionless:** Market making is capital-intensive and requires sophisticated infrastructure.
- **Limited to liquid assets:** Niche outcomes cannot sustain a continuous order book.

1.2 The Parimutuel Gap

Despite parimutuel betting being a centuries-old, mathematically sound mechanism — and the dominant model for horse racing, greyhound racing, and jai alai worldwide — it has not been systematically applied to decentralized prediction markets. The reasons are primarily historical, not technical:

- Lack of blockchain infrastructure for trustless pool management
- Over-reliance on AMM models during the DeFi boom
- Focus on continuous trading rather than event-driven markets

Parl fills this gap.

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2. The Parimutuel Mechanism

2.1 How Parimutuel Betting Works

Parimutuel betting is a pool-based wagering system. Unlike fixed-odds betting (where odds are set by a bookmaker), parimutuel odds are determined collectively by all bettors.

```
\begin{figure}[h]
\centering
\begin{tabular}{|l|c|c|}
\hline
\textbf{Step} & \textbf{Action} & \textbf{Result} \\ \hline
1. Bettors place bets on outcomes A, B, C & Contributions added to single pool & 
2. Event resolves & Oracle or dispute mechanism reports outcome & Winning pool determined
3. Protocol deducts fee (vig) & Typically 2--5\% & House revenue collected
4. Winners split remaining pool & Payout = (Pool  $\times$  (1 - vig)) / #winners
\end{tabular}
\caption{Parl: Parimutuel lifecycle}
\end{figure}
```

2.2 Mathematical Model

Let:

- P = total pool (sum of all bets)
- v = vig (protocol fee, expressed as decimal, e.g., 0.03 for 3%)
- W = total amount bet on winning outcome(s)
- B_i = individual bet amount on winning outcome

Net pool after fee:

$$P_{\text{net}} = P \times (1 - v)$$

Payout per unit bet on winning outcome:

$$\text{Odds}_{\text{winning}} = \frac{P_{\text{net}}}{W}$$

Individual payout:

$$\text{Payout}_i = B_i \times \frac{P_{\text{net}}}{W}$$

Implied probability (before outcome):

For outcome j with total bets B_j :

$$\text{Prob}_j = \frac{B_j}{P} \times (1 - v)$$

2.3 Key Properties

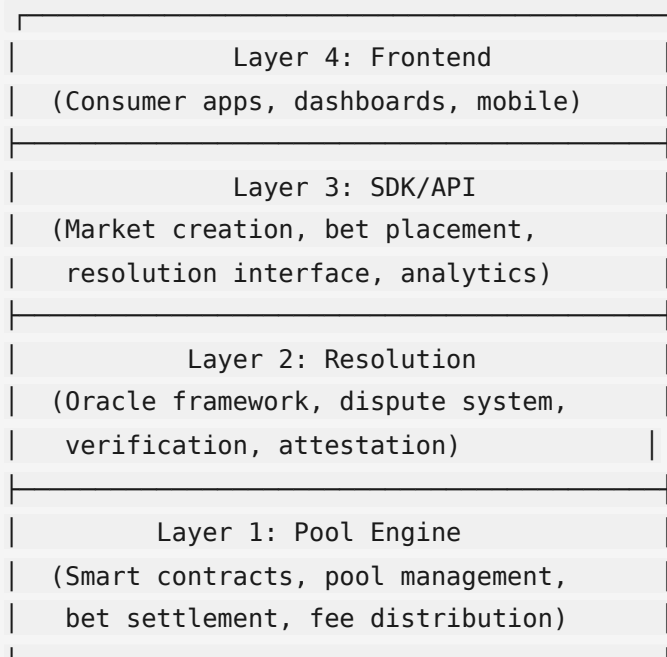
Property	Implication
Zero liquidity risk	No LPs, no IL, no impermanent loss. The pool is self-contained.
Infinite scalability	Any number of outcomes can be added. Pool grows with participation.
Market-maker-free	No continuous quoting required. Odds emerge naturally from bet distribution.
Capped downside	Protocol only pays out what is in the pool. No solvency risk.
Simple math	Single formula applies to all markets. Easily auditable.

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3. Protocol Architecture

3.1 Layer Model

Parl is designed as a four-layer protocol stack, following infrastructure-first principles.



Layer 1 — Pool Engine

The core smart contract layer. Responsibilities:

- Accept bets into named market pools
- Track individual bet amounts and outcomes
- Calculate and distribute payouts after resolution
- Collect protocol fees
- Prevent double-claiming and replay attacks

Layer 2 — Resolution Framework

An oracle-agnostic resolution layer that markets can configure:

- **Built-in adapters:** API-based oracles (e.g., news data feeds)
- **Optimistic resolution:** Bonded proposers with dispute windows
- **Decentralized arbitration:** Integration with Kleros, Reality.eth, or custom dispute boards
- **Community vote:** Token-weighted or reputation-based resolution for informal markets

Markets choose their resolution mechanism at creation time.

Layer 3 — SDK / API

Developer-facing tooling for:

- Creating markets (outcomes, parameters, fee structure, resolution method)
- Placing bets and checking positions
- Querying pool state and odds
- Resolving and settling markets
- Webhook notifications for market lifecycle events

Layer 4 — Frontend (Partner/Community)

Parl does not build its own consumer frontend initially. Instead:

- Partners build branded applications on top of the SDK
- A reference implementation (open-source UI) will be provided for community deployment
- Protocol-owned frontend optional at later stage

3.2 Smart Contract Design (Conceptual)

Core Contracts

```
PoolFactory
├─ createMarket(params) → Market
└─ getPools(filter) → Market[]
```

```

Market
├─ placeBet(outcome, amount) → Receipt
├─ resolve(outcome, proof) → void
├─ claim(winningAddress) → Payout
├─ getState() → PoolState
└─ getOdds() → OddsSnapshot

FeeModule
├─ setVig(marketId, basisPoints) → void
├─ accumulateFees() → uint256
└─ withdrawFees(recipient) → void

ResolutionOracle (interface)
├─ proposeOutcome(marketId, outcome) → void
├─ dispute(marketId, evidence) → void
└─ finalize(marketId) → Outcome

```

Key Design Decisions

Decision	Rationale
Market-level fee configuration	Allows partners to set their own vig within protocol minimum/maximum
Outcome-based receipts	Non-transferable receipts stored as mapping; prevents secondary market complexity
Time-locked resolution	Minimum N blocks between outcome proposal and settlement for dispute window
Pausable markets	Emergency pause for oracle disputes or protocol upgrades
Upgradeable via proxy	Market logic can evolve via UUPS proxy pattern

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4. Resolution & Oracles

4.1 The Oracle Problem

The most significant technical challenge for any prediction market — including Parl — is outcome resolution. The protocol must determine, in a cryptographically verifiable way, which outcome actually occurred.

Parl does not mandate a single oracle. Instead, it provides a **resolution framework** that market creators can configure.

4.2 Resolution Methods

4.2.1 Centralized Oracle (Phase 1)

Suitable for initial launch and curated markets.

- Protocol operators (or partners) sign outcome attestations
- Public verifiability via ECDSA signatures on-chain
- Dispute window (e.g., 24-48 hours) for users to challenge
- Fast, low cost, simple UX

Best for: Sports scores, election results, price feeds, curated event markets.

4.2.2 Optimistic Oracle (Phase 2)

Inspired by UMA's Optimistic Oracle.

- Any bonded proposer can submit an outcome
- Bond is at risk during dispute window
- Disputers can challenge by posting their own bond
- Dispute resolved via decentralized arbitration
- Honest proposers are rewarded; dishonest ones are slashed

Best for: Permissionless market creation where operator curation is undesirable.

4.2.3 Decentralized Arbitration (Phase 2+)

- Integration with established dispute resolution protocols (Kleros, Aragon Court)
- Jury-based or game-theoretic resolution
- Higher cost but maximum trustlessness

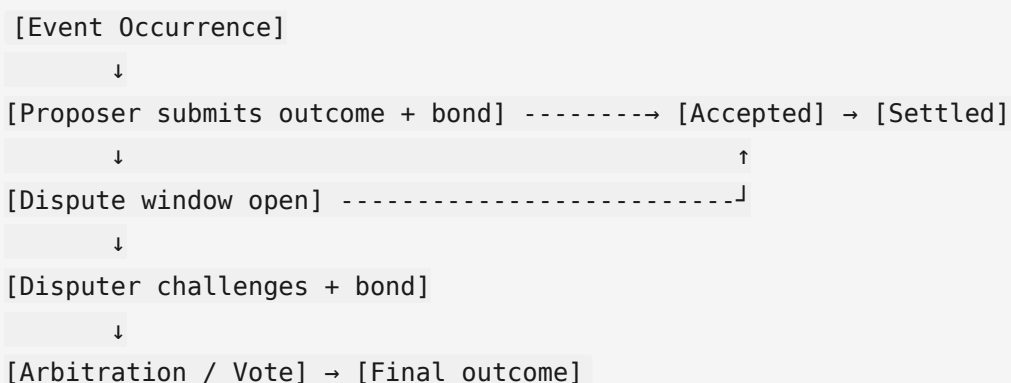
Best for: High-value markets, contentious outcomes.

4.2.4 Community Vote (Phase 2+)

- Token-weighted or reputation-weighted voting
- Suitable for subjective outcomes ("Best movie of 2026")
- Requires anti-sybil mechanisms

Best for: Informal markets, opinion-based predictions, community engagement.

4.3 Resolution Lifecycle



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5. Token Economics

5.1 Protocol Value Accrual

The Parl protocol is designed to be token-agnostic at the core — any asset can serve as a pool's denomination. However, protocol \$PARL token provides:

5.1.1 Fee Discounts

Market creators and bettors who stake/hold \$PARL receive reduced protocol fees.

Tier	\$PARL Staked	Fee Discount
Bronze	1,000+	10%
Silver	10,000+	25%
Gold	100,000+	50%

5.1.2 Governance

- Protocol fee parameters
- Treasury allocation
- Oracle whitelist (Phase 2)
- Protocol upgrades

5.1.3 Revenue Streams

Source	Share
Protocol vig (per-pool)	2--5% of pool
Partner SDK licensing	Tiered fee
Premium oracle service fee	Per-market fee

5.2 Fee Structure

Parl charges a **protocol fee** (vig) on each market's pool. This fee is configurable per-market within a protocol-defined range (e.g., 1--5%).

- **Minimum fee:** 1% — ensures protocol sustainability
- **Default fee:** 3% — competitive with sportsbook vig (typically 5--10%)

- **Custom fee:** Partners can set higher vig; surplus shared via revenue split

5.3 Treasury

Protocol fees accumulate in a treasury governed by \$PARL holders:

- **50%** — Protocol development and security
 - **25%** — Partner incentives and liquidity bootstrapping
 - **15%** — Community grants and hackathons
 - **10%** — Reserve
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6. Product Roadmap

6.1 Phase 0 — Foundation (Q3 2026)

- Core pool engine smart contracts (EVM-compatible)
- Centralized oracle adapter
- Basic SDK (market creation, bet placement, claim)
- Reference frontend (open-source, minimal)

6.2 Phase 1 — Launch (Q4 2026)

- Testnet deployment
- Security audit (two independent firms)
- Partner onboarding program
- Initial curated markets (sports, crypto events)
- \$PARL token launch

6.3 Phase 2 — Expansion (Q1 2027)

- Optimistic oracle integration

- Multi-chain deployment
- Mobile SDK (React Native)
- Partner-facing dashboard
- Community market creation

6.4 Phase 3 — Ecosystem (Q2 2027+)

- Decentralized arbitration integration
- Cross-chain pool interoperability
- Enterprise-grade API (SLAs, dedicated support)
- DAO governance launch
- Parl Improvement Proposal (PIP) process

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7. Competitive Landscape

Feature	Parl	Polymarket	Augur	Azuro
Mechanism	Parimutuel	AMM / CLOB	AMM	AMM
Market Maker	Not required	Required	Required	Required
Liquidity Risk	None	IL for LPs	IL for LPs	IL for LPs
Scalability	Infinite	Per-pool bounded	Per-pool bounded	Per-pool bounded
Oracle Model	Agnostic	Custom + UMA	Reality.eth	Oracle.network
Fee Flexibility	Per-market config	Fixed	Fixed	Fixed
Permissionless	Phase 2	Yes	Yes	Limited

7.1 Moats

1. **Network effects of pool diversity:** Every market adds value to the platform, and no market is too niche to list.
 2. **Partner ecosystem:** SDK-first approach allows Parl to power hundreds of branded prediction platforms.
 3. **Data flywheel:** Resolution track record builds trust; trusted resolution enables more markets.
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8. Risk Factors

8.1 Regulatory Risk

Prediction markets operate in a legally gray area in many jurisdictions. Parl mitigates this by:

- Being infrastructure-first (protocol/SDK, not a consumer betting platform)
- Allowing partners to handle compliance in their jurisdictions
- Offering permissionless deployment (no central operator control post-launch)
- Legal review of tokenomics to avoid securities classification

8.2 Oracle Risk

The weakest link in any prediction market. Mitigations:

- Multiple resolution methods per market
- Bonded dispute mechanism for economic security
- Graduated decentralization (start curated, open over time)
- Explicit disclaimers and market-specific oracle documentation

8.3 Smart Contract Risk

- Two independent security audits before mainnet

- Bug bounty program (\$500k+)
- Gradual TVL caps during launch phase
- Emergency pause mechanism
- Formal verification for core pool math

8.4 Adoption Risk

Prediction markets remain niche. Mitigations:

- Focus on use cases with proven demand (sports, crypto events)
 - Low barrier to entry for market creators
 - Partnership strategy over direct-to-consumer
 - Emphasis on markets that cannot exist on traditional platforms (community, hyper-niche)
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9. Conclusion

Parimutuel mechanics are mathematically elegant, economically sound, and battle-tested over centuries of horse racing. Their application to decentralized prediction markets unlocks the one property that existing solutions cannot offer: **infinite scalability without liquidity burden.**

Parl is not just another prediction market. It is a protocol layer — a piece of infrastructure that any application, community, or enterprise can use to create transparent, trust-minimized markets on any topic.

The timing is right. Blockchain infrastructure has matured. Oracle technology is viable. The demand for alternative betting and forecasting platforms is growing. What has been missing is a mechanism that makes prediction markets as easy to launch as a group chat — without worrying about who will make the market, provide the liquidity, or stabilize the price.

Parl solves that.

The next step is building the pool engine, proving the concept with real markets, and growing an ecosystem of partners who build on top.

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Appendix A: Glossary

Term	Definition
Vig (vigorish)	The protocol's commission on each pool.
Pool	Total amount of all bets on a single market.
Parimutuel	A betting system where all bets on all outcomes are pooled, and winners share the pool proportionally.
AMM	Automated Market Maker — a smart contract that provides liquidity algorithmically.
IL	Impermanent Loss — loss LPs incur when asset prices diverge from deposit ratio.
Oracle	A system that reports real-world data to the blockchain.
Optimistic Oracle	An oracle where data is assumed correct unless challenged within a dispute window.
Slippage	The difference between expected and executed price due to insufficient liquidity.
CLOB	Central Limit Order Book — a system that matches buy and sell orders.

Appendix B: Comparison of Betting Mechanisms

	Fixed Odds	AMM	Parimutuel (Parl)
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Risk borne by	Bookmaker	LP	No one (pool is self-contained)
Liquidity required	Large	Large	Zero (grows with bets)
Odds stability	Fixed at placement	Variable (slippage)	Variable (until outcome)
Scalability	Linear with bookmaker capital	Per-pool bounded	Infinite
Mathematical complexity	Low	Medium	Low
Transparency	Opaque	On-chain	On-chain

This document is a draft whitepaper for informational purposes only. It does not constitute a solicitation to purchase tokens or securities. Parl is a work in progress and specifications may change.